

Causal Inference for Intervention & Service Evaluations: A Practical Handbook

Jamie Wong

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Welcome

This is a practical handbook on causal inference methods for evaluating policy interventions and service changes within the NHS, with a focus on leveraging observational real-world data to generate robust evidence.

The handbook was written initially as part of an NHS England PhD Internship done jointly with the NHS North Central London Integrated Care Board and University College London, and will continue to be periodically updated with revisions following feedback from academics, data scientists, data analysts, clinicians, and public health practitioners.



Part I

Introduction

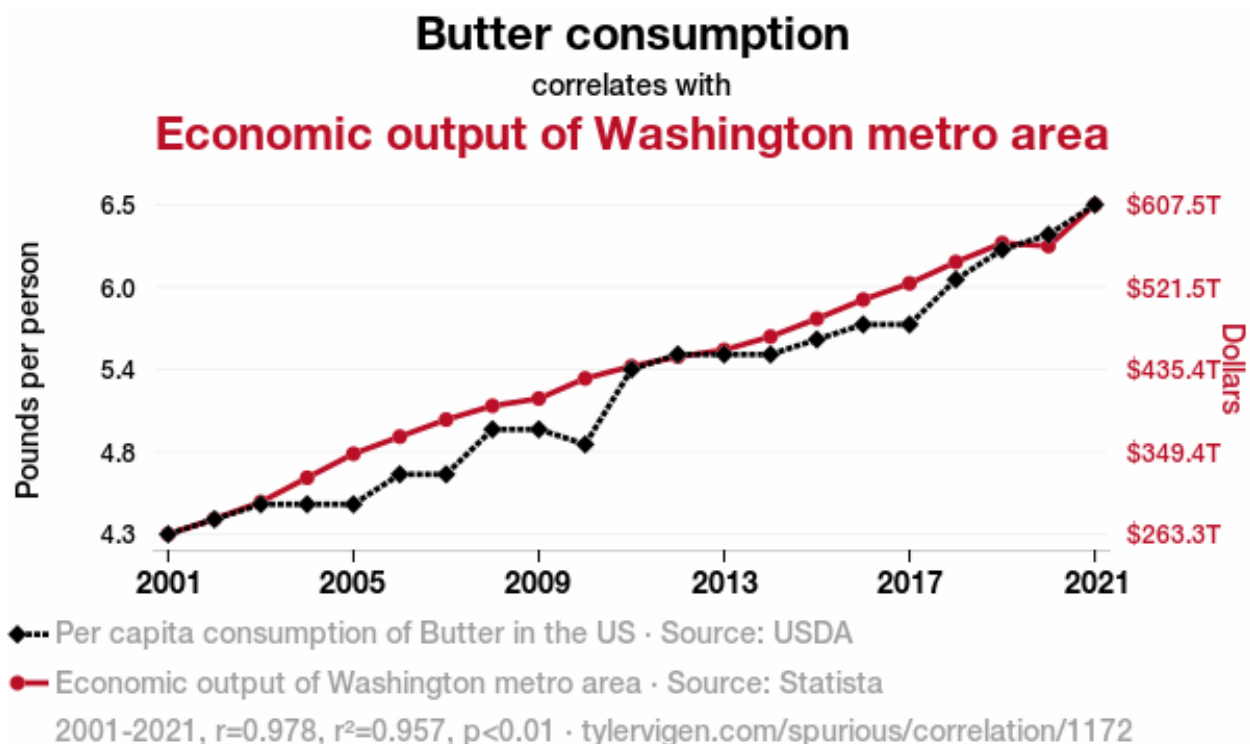
1 Background & Motivation for Causal Inference in Service Evaluations

When performing evaluations of healthcare services or interventions in the NHS, our key evaluation question of interest is almost always causal in nature:

- Has our new breast cancer screening programme reduced mortality?
- Did implementing a community-based model of care divert patients away from A&E services?
- Did implementing a hub and spoke model reduce clinical harm and lower patient waiting times?

Yet many of the methods currently used to evaluate interventions within the NHS are often descriptive in nature. Dashboards are frequently used in this manner, where any changes in an outcome of interest are fully attributed to a single change in implementation in services, without considering whether there are alternative explanations and/or other confounding factors are involved that could instead explain the change instead.

To better illustrate this point, Tyler Vigen publishes great examples of spurious correlations on his webpage (<https://tylervigen.com/spurious-correlations>), from which I've attached an example of below (Vigen 2024):



As you can see, butter consumption is positively correlated with the economic output of the Washington metro area. Needless to say, I highly doubt butter consumption was actually “greasing the wheels” of the local economy in the Washington metro area! It is therefore imperative that we don’t conflate association with causation when we’re conducting service evaluation as well.

Many books, papers, and reports have been published providing guidance on how one can estimate the causal effect of different interventions have on specific outcomes, in the absence of a randomised trial. Some resources have even been developed to help narrow down which types of methods would be most appropriate depending on the context. However, much of the guidance has been limited to methods originating from a specific field (e.g., econometrics, social sciences, epidemiology), rather than encompassing all approaches relevant to service and intervention evaluation within the realm of health and social care.

This handbook aims to bridge this gap in the literature, and establish new unifying guidance for selecting appropriate study designs and analytical methodologies for conducting causal inference in service evaluations. You’ll find:

1. A decision tree diagram to aid with method selection
2. Concise overviews of each approach
3. Pointers to more in-depth resources for technical details about each method (including technical explanations in textbooks, applied examples in the literature, and practical coding exercises)

Do note that this handbook itself will not be going into too much technical detail about each analytical method, but instead serves as a resource to guide analysts in choosing the appropriate to use for their question.

I would also like to note that this handbook will primarily focus largely on established statistical methods for causal inference. More novel and experimental machine learning based methods do exist for causal inference. However, most of these methods require additional computational power, have components that are more opaque (akin to a black box), and many of these still haven’t been extensively used in the literature yet. For the purposes of keeping this guide as approachable as possible, and our methods and assumptions as transparent as we can, this is the focus we have chosen for this handbook. An extensions section will be included at the end for those who may be interested in further delving into novel machine methods as well.

2 A Brief History on Conceptualising Causality

2.1 The Potential Outcomes Framework

The field of causal inference has a long history built on insights from multiple disciplines. Multiple schools of thought have been developed, contributing different ideas on how to conceptualise and define the concept of causality.

One of the most widely adopted ways to conceptualise causal inference in the fields of applied statistics, econometrics, and epidemiology is through the idea of potential outcomes, also referred to as the counterfactual approach to causality. This is where we aim to determine if the outcome for an individual would have been different if given different hypothetical value(s) of the exposure of interest.

Such counterfactual ideas stretch back to key texts written during the early to mid 19th century:

“If a person eats of a particular dish, and dies in consequence, that is, would not have died if he had not eaten of it, people would be apt to say that eating of that dish was the cause of his death” (Mill 1843).

However, it wasn’t until Neyman’s 1923 Master’s thesis where the notation for potential outcomes was first formalised, introducing the idea that within randomized experiments, if we’re interested in estimating the effect of a treatment or intervention, each unit in a study for a binary outcome can only have two fixed potential outcomes, one under treatment and one under control (Splawa-Neyman, Dabrowska, and Speed 1990).

In 1925, Fisher proposed randomizing treatments to study units/participants in order to derive unbiased inferences from experiments, though without reference to potential outcomes or estimating average treatment effects (Ronald A. Fisher 1925; R. A. Fisher and Yates 1990).

This changed in 1974, where Rubin formally established what we now refer to as the potential outcomes framework, extending these ideas to non-randomized observational studies, and writing formal notation for calculating the average casual effect that is used today (Rubin 1974). Holland later named this the Rubin Causal Model, and emphasised the fundamental problem of causal inference:

“It is impossible to observe [the value of the response that would be observed if the unit was exposure to treatment] and [the value that would be observed on the same unit if it were exposed to the control], and therefore it is impossible to observe the effect of [treatment] on [the unit of interest]” (Holland, Glymour, and Granger 1985).

In short, if we’re interested in evaluating a treatment, one cannot observe both hypothetical outcomes for a single unit or individual, one in a universe where the unit/individual of interest is treated, and another where the same unit/individual is not treated.

Because the potential outcomes framework mirrors the logic found in randomized trials, the school of thought has underpinned virtually every method covered in this handbook, including those with origins in econometrics and epidemiology.

2.2 Structural Causal Models and Directed Acyclic Graphs

However, alongside the counterfactual school of thought, competing theories were developed in parallel, the most prominent of which came from Pearl. Rather than framing causation in terms of counterfactual outcomes, Pearl popularised the development of Structural Causal Models (SCMs), which use Directed Acyclic Graphs (DAGs) and structural equations to present causal relationships and underlying assumptions in both a visual and mathematical way (Pearl 2009; Pearl and Mackenzie 2018).

I will be delving into the specifics of Directed Acyclic Graphs in their own section later into this handbook. As a brief overview though, DAGs are diagrams which include each of your variables (exposure, covariates, and outcomes) as nodes, connected by arrows that indicate a direct causal relationship between them. Digitale et al., from UCSF have a great tutorial on DAGs if you'd like a brief introduction on the topic (Digitale, Martin, and Glymour 2022).

Historically when these methods were introduced, the two schools of thought were completely split, with econometricians favouring the potential outcomes approach, while computer scientists favoured structural causal models. A great discussion on this is provided by Nobel laureate Imbens (an economist) more recently, who argues that although these two frameworks are complementary, the potential outcomes approach is often preferred for econometrics because it aligns naturally with study designs that exploit policy changes or other “as-if random” interventions, rather than collecting data on and adjusting for a wide range of individual variables mapped out on a DAG (Imbens 2020). This type of thinking was how the field of quasi-experimental methods originated, the specifics of which will be discussed in later sections of the handbook.

Overtime though, some fields such as Epidemiology and Public Health have found ways to combine both school of thought into a single workflow:

1. DAGs are drawn to visualise causal relationships, identify confounders of interest, present assumptions made transparently.
2. Use back door criteria on the DAG to select the minimal set of covariates that require adjustment of to make an unbiased estimate of the causal contrast of interest (e.g., average treatment effect, average treatment effect on the treated).
3. Apply appropriate statistical methods to adjust for the set of covariates, such as regression modelling, to estimate the target causal contrast, under the potential outcomes framework. This hybrid approach works particularly well in settings with clearly defined treatment contrasts (say, drug X vs. drug Y) and rich data from longitudinal studies and electronic health records, which are harder to come by in economics.

This hybrid approach works particularly well in settings with clearly defined treatment contrasts (say, drug X vs. drug Y) and rich data from longitudinal studies and electronic health records, which are harder to come by in economics.

If you are interested in reading further into the origins of the potential outcome framework and quasi-experimental designs, Cunningham’s supplementary teaching material hosted on github as part of his “Causal Inference: The Mixtape” book goes into this in more depth (Cunningham 2025). His slides of the “[Foundation of Causality](#)” in particular focuses on the history of these foundational ideas. The introductory chapter of Morgan and Winships’ “Counterfactuals and Causal Inference,” as well as chapter 2 of Imbens and Rubins’ “Causal Inference for Statistics, Social, and Biomedical Sciences” also provide a great overview of the historical context behind how causal inference as we know it today was first formalised (Morgan and Winship 2014; Imbens and Rubin 2015).

2.3 Additional Approaches to Establishing Causality

During the mid-19th century, before Rubin had established the potential outcomes framework, Hill had notably published a seminal paper covering a set of criteria that would help establish causality within epidemiological studies, now commonly referred to as the “Bradford Hill Criteria.” These included: strength, consistency, specificity, temporality, biological gradient, biological plausibility, coherence, experimental evidence, and analogy (Hill 1965). The criteria, however, did not provide a practical way for epidemiologists and social scientists more generally to design observational studies targeting causal inference though. The criteria is therefore used more so as a guide for evaluating whether associations are causal while evaluating a set of evidence, and still often used in policy evaluations. Chapter 2 of Lash et al.’s textbook “Modern Epidemiology” provides a great explanation of each criteria if you’d like to delve deeper into this.

Additional causal frameworks have been built on top of this foundation as well, namely the Target Trial Emulation framework by Hernan and Robins which will be discussed in further detail in its own section (Hernán and Robins 2016). The framework aims to overcome many of the methodological issues facing classical epidemiological analyses by framing causal questions of interest within an ideal pragmatic randomized trial. This will underpin much of the methods covered as part of the “Epidemiology Methods” section further into the handbook.

3 Common Assumptions in Causal Inference Methodology

Before delving into individual methods, here's a short reference of common assumptions that will be referred to throughout this handbook. Not every method relies on all of these assumptions, but this will serve as your go to reference point in case there is any confusion between them later on. Method specific assumptions will be detailed within each methods section as well.

3.1 Exchangeability

Markozannes et al., define exchangeability as holding when: *“Treatment assignment is independent of the potential outcomes; this roughly translates to no unmeasured confounders and no informative censoring”* (Markozannes, Vourli, and Ntzani 2021).

Exchangeability can be divided into two types:

1. Unconditional exchangeability
 - Where in a randomized trial, potential outcomes are inherently independent from treatment assignment due to random assignment of treatment.
2. Conditional exchangeability
 - Where in an observational study, potential outcomes are only independent from treatment assignment given/controlling for a set of measured covariates.

Hernan and Robins' definition is framed more as conditional exchangeability in their book, given its focus on observational studies, and define exchangeability as holding when: *“The conditional probability of receiving every value of treatment, though not decided by the investigators, depends only on measured covariates,”* i.e. no residual confounding (Hernán and Robins 2025).

For example, if we're studying the relationship between smoking and cardiovascular disease, and sex is known to be independently associated between the two variables and does not lie on the causal pathway (i.e. a known confounder), we would violate the condition by not adjusting for sex in our analysis.

Exchangeability is also referred to as ignorability (ignorable treatment assignment mechanism) or unconfoundedness in the literature.

3.2 Consistency

Hernan and Robins define consistency as holding when: “The values of treatment under comparison correspond to well-defined interventions that, in turn, correspond to the versions of treatment in the data” (Hernán and Robins 2025).

For example, in a study evaluating the effect of paracetamol on chronic pain, if some participants have one dose a week, and others have one daily, that would violate the condition, as the intervention is not uniformly defined.

3.3 Positivity

Hernan and Robins define positivity as holding when: “The probability of receiving every value of treatment conditional on relevant covariants is greater than zero, i.e., positive” (Hernán and Robins 2025).

For example, if we’re studying the effect of physical activity on the incidence of obesity, participants who are disabled and bedbound have a 0% chance of receiving the exercise intervention, thereby violating the condition.

3.4 Stable Unit Treatment Value Assumption (SUTVA)

The Stable Unit Treatment Value Assumption (SUTVA), as defined by Rubin who coined the term: *“Is simply the a priori assumption that the value of Y for unit u when exposed to treatment t will be the same no matter what mechanism is used to assign treatment t to unit u and no matter what treatments the other units receive”* (Rubin 1986).

A simpler definition and explanation has been published by Markozannes et al.: *“The stable unit treatment value assumption states that there is no interference among units, that is, the treatment status of a unit does not affect the potential outcomes of other units and it also requires that there is only a single version of the treatment (no hidden variations in treatment; no multiple versions of treatment).”* *“Possible violations of the SUTVA include settings where units interact (e.g., schools, group interventions) or different treatment dosages exist or different modes of administration operate which can affect the potential outcomes”* (Markozannes, Vourli, and Ntzani 2021).

In other words, we can define SUTVA has having two distinct requirements:

1. No multiple versions of treatment
2. No interference between units

A more concrete example would be in the context of evaluating the efficacy of a new flu vaccine, where the assumption would hold as long as:

1. All versions of the flu vaccines are the same, and are administered in the exact same manner between study participants (i.e., have the same dosage, formulation, and schedule)
2. Each study participant is isolated in separate facilities, with limited contact with other trial participants and non-participants

Of course, in practice SUTVA would not hold in an evaluation of a flu vaccine, as study participants would come into contact with other trial participants and non-participants, introducing herd level spillover effects into the population. Do note though that defining the unit of interest differently could change this depending on conditions (e.g., going from an individual level analysis to a city or country level analysis).

SUTVA is also referred to as a non-macro-effect or the partial equilibrium assumption in economics (Morgan and Winship 2014).

Assumptions made when using some of the most common statistical techniques discussed in this handbook can be found in this [paper](#).

4 Existing Guidance on Study Design Selection

Prior to presenting my unifying guidance in my next section, I've attached a few existing reports and guides for selecting the appropriate observational causal study design relevant to your key evaluation question of interest. This was done to acknowledge the work that has come before, while also noting key limitations that motivated the development of this unifying guidance.

4.1 NICE Guidance

In their report on “methods for the development of NICE public health guidelines,” the National Institute for Health and Care Excellence (NICE) provides a high-level overview of the various epidemiological study designs used in public health research. Included within the report in Appendix E is a flowchart that outlines an “algorithm for classifying quantitative (experimental and observational) study designs (NICE 2012).

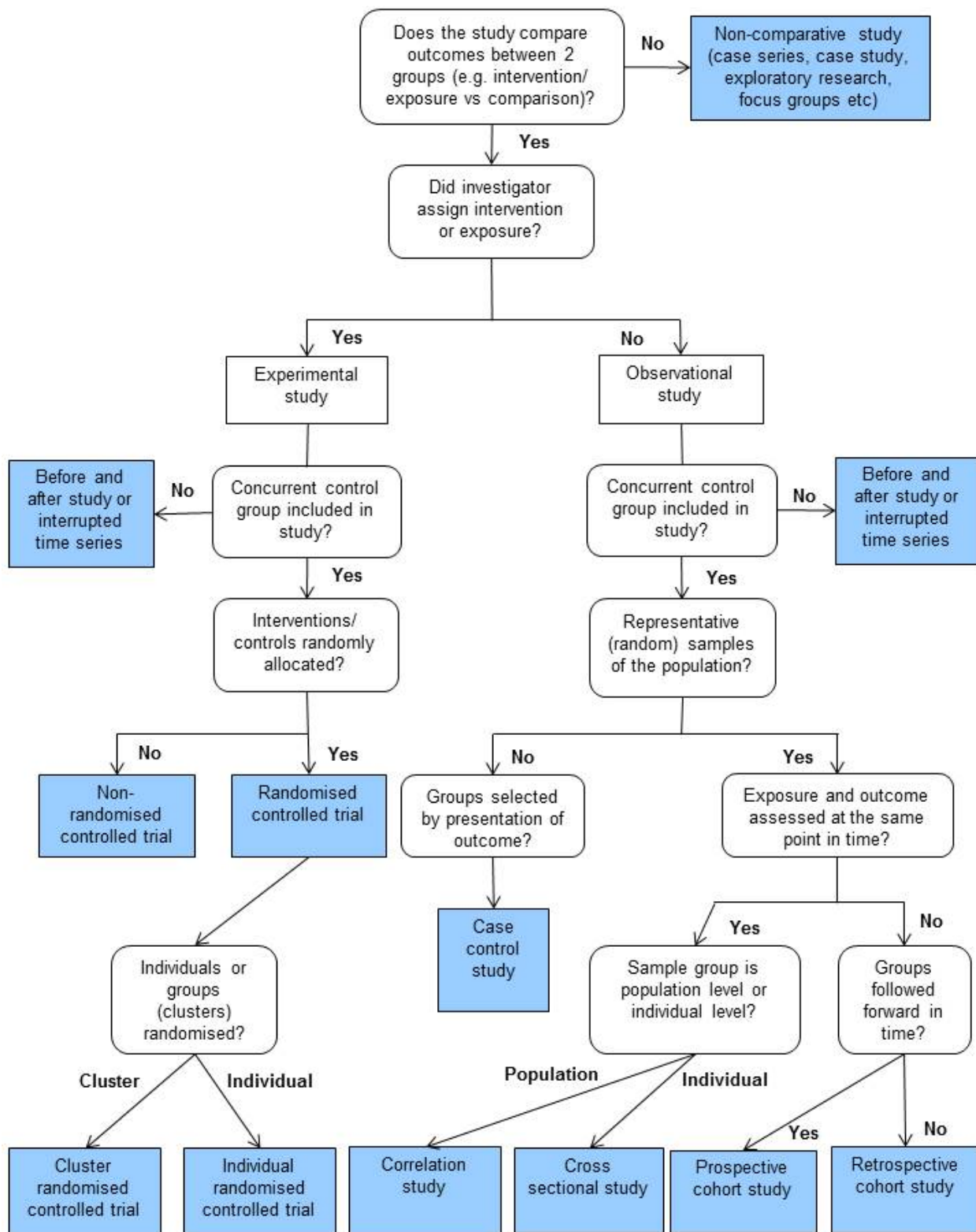


Figure 4.1: Algorithm for classifying quantitative (experimental and observational) study designs (NICE, 2012)

As you can see, the types of study designs covered are very general, and do not focus on specific types of analytical methods you can use to deal with confounding factors within your study to derive causal inferences. As such, this diagram is most useful for those who are new to the field of epidemiology and public health, and wish to refer to a general overview of experimental and observational study designs.

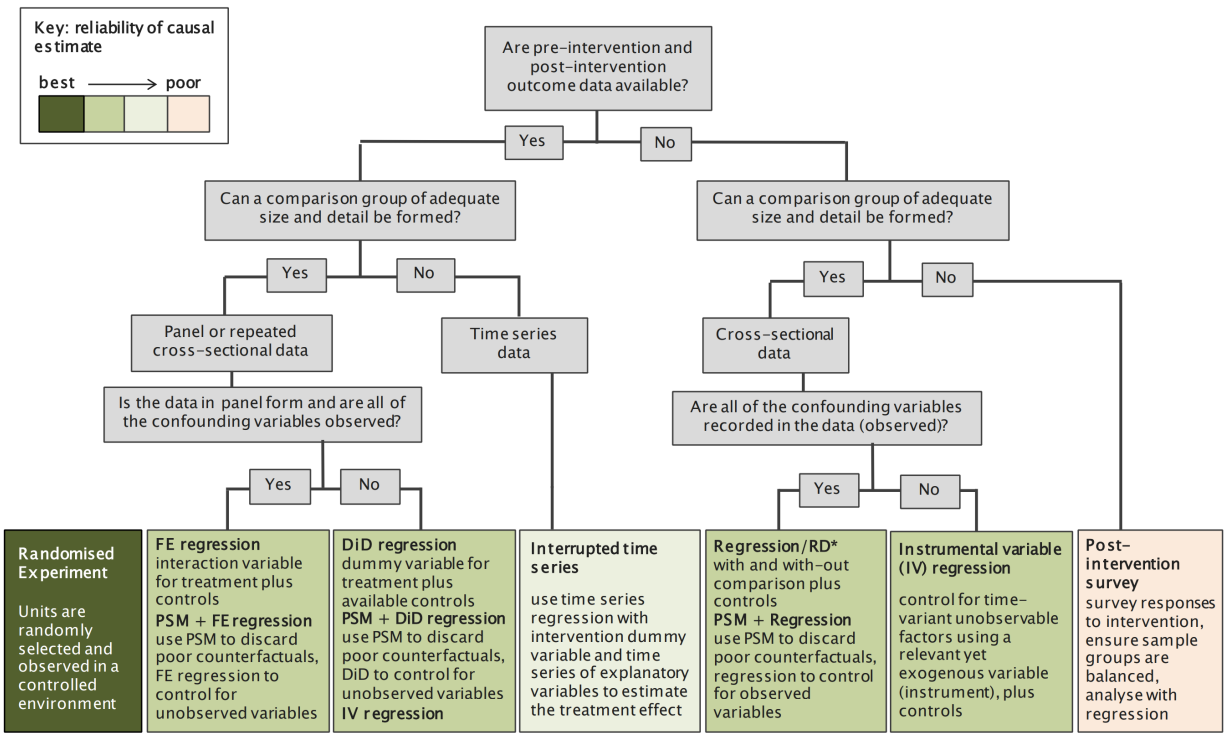
4.2 OHID Guidance

The Office for Health Improvement and Disparities (OHID) has similarly published some high-level guidance on the use of various evaluation methods, including quasi-experimental methods, to evaluate digital health products (OHID 2021a, 2021b). This guidance is aimed at analysts that are completely new to these methods though, and as such does not go into the specifics of how to perform these analyses in a programming language and/or statistical package such as R, Stata, and Python.

4.3 New Zealand Transport Agency Guidance

The New Zealand Transport Agency Waka Kotahi has published a report focused on the use of causal inference methods for the evaluation of transport interventions, and includes a causal inference method selection flowchart (Schiff, Wright, and Denne 2017).

The methods included are limited to those most often used in the social sciences and econometrics though. These methods largely exploit natural experiments that have occurred in the real world at one point in time to derive causal inferences. There is hence room to expand upon this flowchart further in a more comprehensive manner.



* Regression Discontinuity (RD) is a special type of cross section regression used when there is no overlap between treatment and control groups.

Figure 4.2: Causal inference method selection flowchart (Schiff, Wright and Denne, 2017)

4.4 HM Treasury Guidance

The HM Treasury's Magenta Book, specifically Figure 3.1 on p.47 (PDF p.54) provides a great diagram providing guidance on when using different experimental and quasi-experimental methods for service evaluation would be the most appropriate (HM Treasury 2020). Unlike the New Zealand Transport Agency's guidance, the Treasury's diagram is more descriptive in nature, describing specific contexts where each approach would be most applicable.

However, similar to the New Zealand Transport Agency's guidance, the quasi-experimental methods presented only encompass methods used more often in econometrics and social sciences, such as difference in differences and regression discontinuity.

These methods once again largely exploit natural experiments that have occurred in the real world to derive causal inferences, rather than adjusting for individual covariates which is more popular in epidemiology. Methods focusing on the latter are absent, but this is no surprise due to the Treasury's area of focus. The guidance presented in this handbook therefore expands upon this diagram to also include such methods.

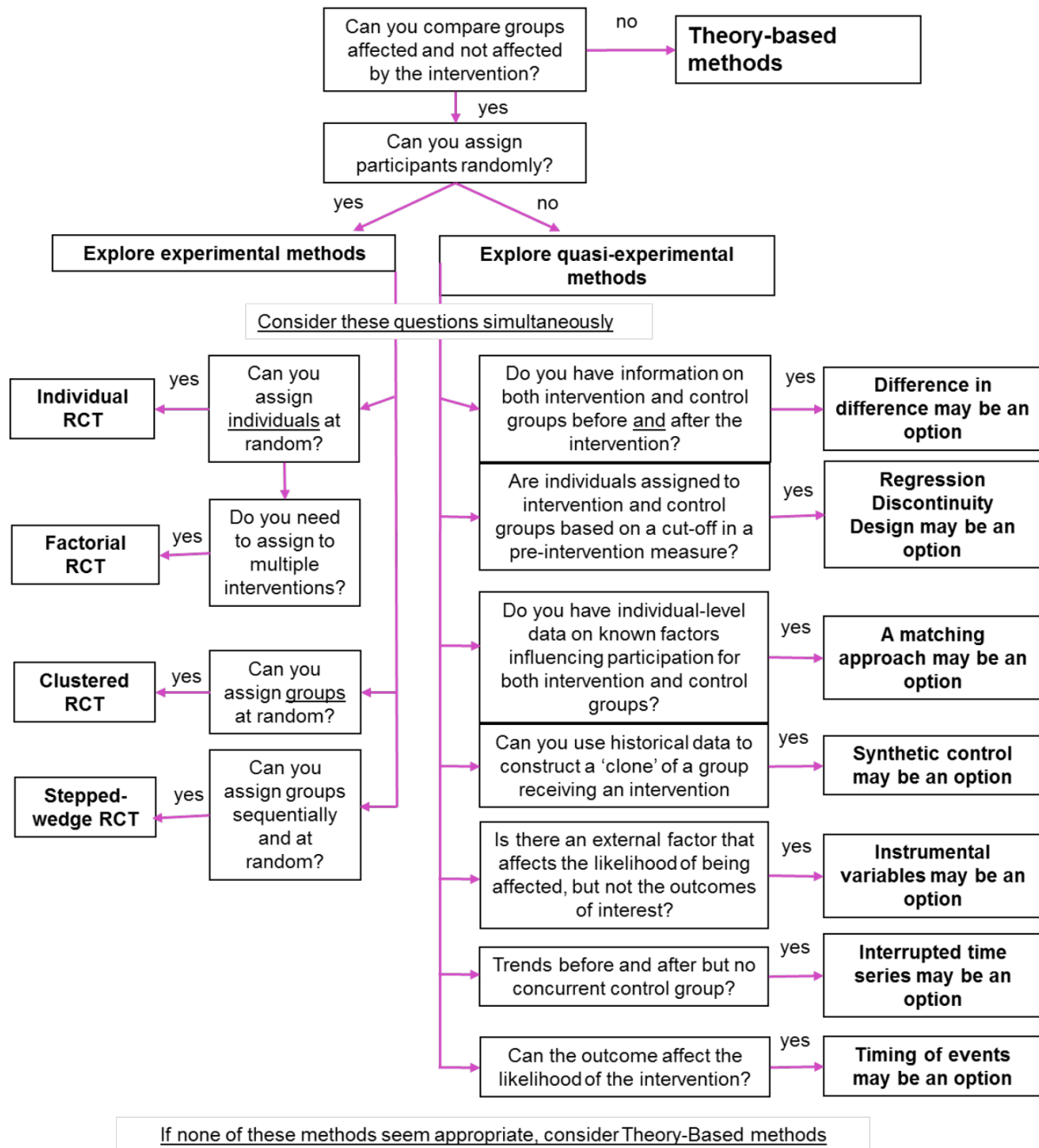


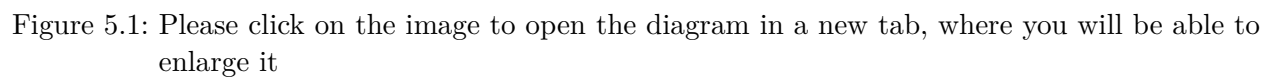
Figure 4.3: Diagram for selecting experimental and quasi-experimental methods (HM Treasury, 2020)

5 New Unifying Guidance on Choosing Causal Inference Methods (Method Selection Tree)

Given the narrow scope of all prior guidance, especially as most of the methods selection diagrams are limited to methods used more within other fields of study, I have opted to create a new method selection tree diagram to help advise with the selection of causal methods going forward.

A flowchart of all methods mentioned in this handbook is presented below, with descriptors indicating which variables are most appropriate given your evaluation question of interest, and the data available to you. Corresponding chapter numbers of each method are also attached within the flowchart.

Please note that the diagram is still a work in progress, and will continue to be updated with input from external experts.



6 Key Resources

Throughout the handbook, there are a few key influential texts that have consolidated contemporary methods used to derive causal inferences from observational data. These texts are widely regarded as being foundational to anyone attempting to dip their toes into the field, and are often used as required reading and/or reference material within undergraduate and postgraduate courses in the econometrics, epidemiology, statistics, and social sciences more broadly.

I've listed some the most frequently cited texts in this handbook below, along with a brief description of their areas of focus. Additional texts that are beyond the scope of this handbook have also been included for reference. A summary table is also included showing which methods are covered by each book in Section 1.7.

Brady Neal has posted a very useful flowchart on his personal website outlining “which causal inference book you should read,” and covers many of the texts mentioned in this handbook (Neal 2019). If you're completely new to the field of causal inference, his diagram serves as a great place to begin your journey.

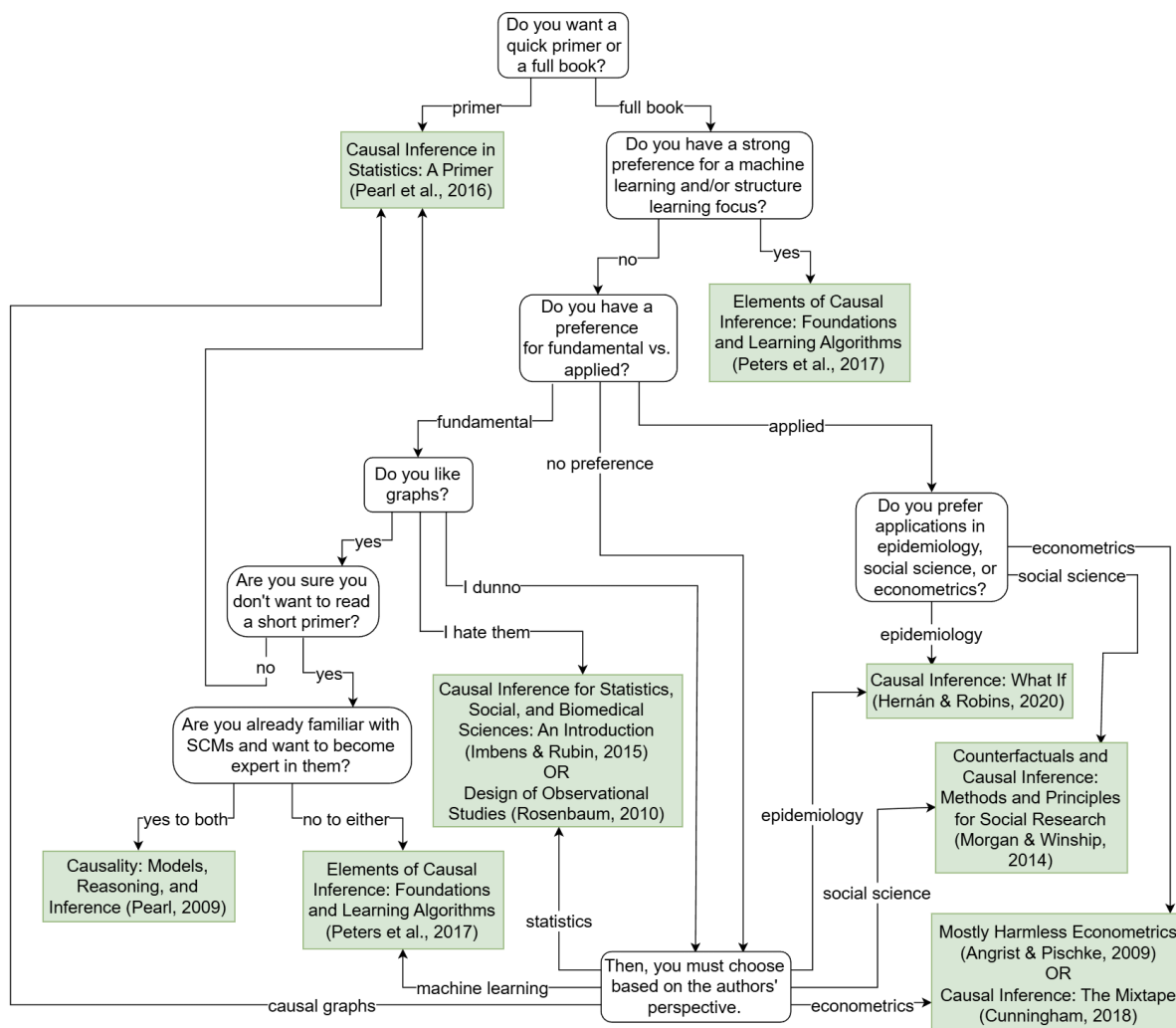
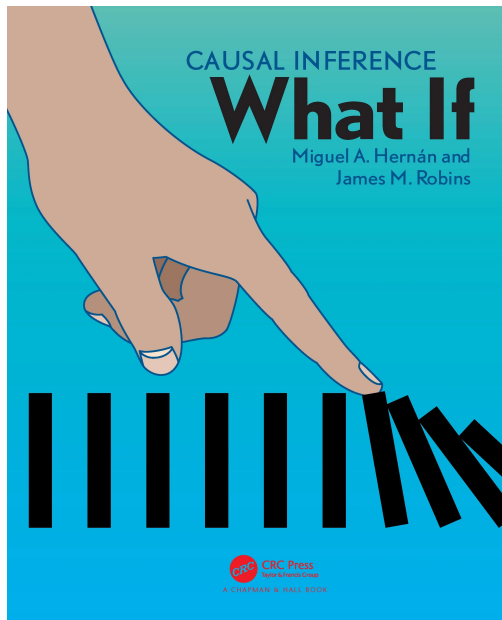


Figure 6.1: Causal Books Flowchart (Neal, 2019)

6.1 Causal Inference: What If (Hernán and Robins 2025)



Authors: Miguel Hernan – *Kolokotronis Professor of Biostatistics and Epidemiology, Harvard T.H. Chan School of Public Health & Director of Harvard CAUSALab* James M. Robins – *Mitchell L. and Robin LaFoley Dong Professor of Epidemiology, Harvard T.H. Chan School of Public Health*

Hernan and Robins' book was first published in 2020, and has been continuously updated over the past few years. The most up to date version of the book can be found on [Hernan's personal webpage](#).

The book is considered by many in the field of epidemiology to be the definitive guide to the application of causal inference methods, framed within the target trial emulation framework, especially when it comes to estimating causal effects using complex longitudinal data (with a strong focus on G-methods). This is a result of the book being openly available online for the past two decades while it was still being drafted, with many researchers in the field contributing to the book over this period.

The book is particularly useful in providing guidance for evaluating sustained and time-varying treatment strategies, and overcoming issues such as immortal time bias within classical methods in epidemiology, which are often issues within the field of pharmacoepidemiology.

The book also (very helpfully) includes accompanying R and Stata code hosted on [GitHub](#), which provides practical examples of how the methods covered can be applied to your own data.

7 Coverage of Causal Inference Methods in Key Texts

Part II

Key Concepts in Causal Inference

8 Randomized Controlled Trials

TBC

9 Directed Acyclic Graphs

TBC

10 Target Trial Emulation

TBC

11 Defining Treatment Strategies

TBC

Part III

Quasi-Experimental Methods

TBC

12 Difference-in-Differences

TBC

13 Interrupted Time Series

TBC

13.1 Standard Interrupted Time Series Analysis

13.2 Controlled Interrupted Time Series Analysis

14 Synthetic Controls

TBC

15 Regression Discontinuity

TBC

16 Instrumental Variables

TBC

17 Extensions of Quasi-Experimental Designs

TBC

17.1 Negative Controls

17.2 Sensitivity Analysis

Part IV

Adjustment-Based Methods

TBC

18 Outcome Regression (Selection on Observables)

TBC

19 Matching (Exact & Covariate)

TBC

20 Propensity Scores

TBC

20.1 Estimating Propensity Scores

20.2 Propensity Score Matching

21 G-Methods: An Introduction

TBC

22 G-Methods 1: Inverse Probability Weighting

TBC

23 G-Methods 2: Parametric G-Formula

23.1 Parametric G-Formula (Non-Iterative Expectation)

23.2 Iterative Conditional Expectation (ICE) G-Formula

24 Extensions of Adjustment-Based Methods

TBC

Part V

Applying Causal Methods: Worked Examples in the NHS

TBC

25 Case Study 1a: DOAC Head-to-Head Emulated Trial

TBC

26 Case Study 1b: DOAC Policy Prescription Change Interrupted Time Series Analyses

TBC

27 # Case Study 2: Ophthalmology Hub of Care Model Interrupted Time Series Analyses

TBC

28 Practical Barriers to Implementation: Common Challenges in Causal Inference

TBC

29 Conclusion

TBC

30 About the Author & Acknowledgements

30.1 About Me

Hi, my name is Jamie and I'm an MRC funded PhD student at UCL, focusing on the application of causal inference methods within Epidemiology, Data Science, and Health Economics.

Before my PhD, I previously graduated with a BSc in Population Health Sciences (Data Science) from UCL, and an NIHR funded MSc in Epidemiology from the London School of Hygiene and Tropical Medicine.

Prior to my PhD, my interest in researching the application of causal inference methods in observational data stemmed from my varied experience working across multiple areas of public health research, including:

- Performing secondary analysis on a nutritional labelling RCT at the NIHR Obesity Policy Research Unit
- Supporting cancer trials as a Clinical Trials Assistant in the MRC Clinical Trials Unit
- Identifying barriers to STI testing at the NIHR Health Protection Research Unit in Blood Borne and Sexually Transmitted Infections
- Quantifying the effects of acute and long-term exposure of air pollutants on respiratory related hospital admissions and related outcomes at UCL
- Evaluating the risk of long-term adverse events in cancer survivors using electronic health record data at LSHTM

Having worked with both randomized trials and observational data, focusing my research on causal inference in real-world data felt like a natural next step.

If you have any questions do feel free to reach out to me via the following channels: UCL Email: [jamie.wong \[at\] ucl.ac.uk](mailto:jamie.wong@ucl.ac.uk) NHS Email: [laichunjamie.wong \[at\] nhs.net](mailto:laichunjamie.wong@nhs.net) LinkedIn: <https://www.linkedin.com/in/jamiewlc/> GitHub: <https://github.com/jamiewlc>

30.2 Acknowledgements

This handbook was written during my PhD placement with NHS England and the North Central London Integrated Care Board as part of the NHS England Data science PhD internship programme, supervised by Jonathan Pearson (NHSE) and Emily Baldwin (NCL ICB).

References

- Cunningham, Scott. 2025. “Mixtape Sessions (Github).” 2025. <https://github.com/Mixtape-Sessions>.
- Digitale, Jean C., Jeffrey N. Martin, and Medellena Maria Glymour. 2022. “Tutorial on Directed Acyclic Graphs.” *Journal of Clinical Epidemiology* 142 (February): 264–67. <https://doi.org/10.1016/j.jclinepi.2021.08.001>.
- Fisher, R A, and F Yates. 1990. *Statistical Methods, Experimental Design, and Scientific Inference: A Re-issue of Statistical Methods for Research Workers, the Design of Experiments and Statistical Methods and Scientific Inference*. Edited by J H Bennett. Oxford University Press. <https://doi.org/10.1093/oso/9780198522294.001.0001>.
- Fisher, Ronald A. 1925. “Statistical Methods for Research Workers (Scan of 1st Edition).” Classics in the History of Psychology. 1925. <https://psychclassics.yorku.ca/Fisher/Methods/>.
- Hernán, Miguel A., and James M. Robins. 2016. “Using Big Data to Emulate a Target Trial When a Randomized Trial Is Not Available.” *American Journal of Epidemiology* 183 (8): 758–64. <https://doi.org/10.1093/aje/kwv254>.
- . 2025. *Causal Inference: What If*. Boca Raton: Chapman & Hall/CRC. <https://miguelhernan.org/whatifbook>.
- Hill, Austin Bradford. 1965. “The Environment and Disease: Association or Causation?” *Proceedings of the Royal Society of Medicine* 58 (5): 295–300. <https://doi.org/10.1177/003591576505800503>.
- HM Treasury. 2020. *Magenta Book: Central Government Guidance on Evaluation*. United Kingdom: HM Treasury. https://assets.publishing.service.gov.uk/media/5e96cab9d3bf7f412b2264b1/HMT_Magenta_Book.pdf.
- Holland, Paul W., Clark Glymour, and Clive Granger. 1985. “Statistics and Causal Inference.” *ETS Research Report Series* 1985 (2): i–72. <https://doi.org/10.1002/j.2330-8516.1985.tb00125.x>.
- Imbens, Guido W. 2020. “Potential Outcome and Directed Acyclic Graph Approaches to Causality: Relevance for Empirical Practice in Economics.” *Journal of Economic Literature* 58 (4): 1129–79. <https://doi.org/10.1257/jel.20191597>.
- Imbens, Guido W., and Donald B. Rubin. 2015. *Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction*. Cambridge: Cambridge University Press. <https://doi.org/10.1017/CBO9781139025751>.
- Markozannes, Georgios, Georgia Vourli, and Evangelia Ntzani. 2021. “A Survey of Methodologies on Causal Inference Methods in Meta-Analyses of Randomized Controlled Trials.” *Systematic Reviews* 10 (1): 170. <https://doi.org/10.1186/s13643-021-01726-1>.
- Mill, John Stuart. 1843. *A System of Logic, Ratiocinative and Inductive: Being a Connected View of the Principles of Evidence, and the Methods of Scientific Investigation*. Vol. 1. Cambridge Library Collection - Philosophy. Cambridge: Cambridge University Press. <https://doi.org/10.1017/CBO9781139149839>.
- Morgan, Stephen L., and Christopher Winship. 2014. *Counterfactuals and Causal Inference: Methods and Principles for Social Research*. 2nd ed. Analytical Methods for Social Research. Cambridge: Cambridge University Press. <https://doi.org/10.1017/CBO9781107587991>.

- Neal, Brady. 2019. “Which Causal Inference Book You Should Read.” November 23, 2019. <https://www.bradyneal.com/which-causal-inference-book>.
- NICE. 2012. “Appendix E Algorithm for Classifying Quantitative (Experimental and Observational) Study Designs | Methods for the Development of NICE Public Health Guidance (Third Edition).” NICE. September 26, 2012. <https://www.nice.org.uk/process/pmg4/chapter/appendix-e-algorithm-for-classifying-quantitative-experimental-and-observational-study-designs>.
- OHID. 2021a. “Guidance: Choose Evaluation Methods: Evaluating Digital Health Products.” GOV.UK. May 19, 2021. <https://www.gov.uk/guidance/choose-evaluation-methods-evaluating-digital-health-products>.
- . 2021b. “Guidance: Quasi-experimental Study: Comparative Studies.” GOV.UK. September 8, 2021. <https://www.gov.uk/guidance/quasi-experimental-study-comparative-studies>.
- Pearl, Judea. 2009. *Causality: Models, Reasoning, and Inference*. 2nd ed. Cambridge: Cambridge University Press. <https://doi.org/10.1017/CBO9780511803161>.
- Pearl, Judea, and Dana Mackenzie. 2018. “The Book of Why: The New Science of Cause and Effect.” Guide books. 2018. <https://dl.acm.org/doi/book/10.5555/3238230>.
- Rubin, Donald B. 1974. “Estimating Causal Effects of Treatments in Randomized and Nonrandomized Studies.” *Journal of Educational Psychology* 66 (5): 688–701. <https://doi.org/10.1037/h0037350>.
- . 1986. “Statistics and Causal Inference: Comment: Which Ifs Have Causal Answers.” *Journal of the American Statistical Association* 81 (396): 961–62. <https://doi.org/10.2307/2289065>.
- Schiff, A, L Wright, and T Denne. 2017. “Ex-Post Evaluation of Transport Interventions Using Causal Inference Methods.” NZ Transport Agency Waka Kotahi. <https://nzta.govt.nz/resources/research/reports/630/>.
- Splawa-Neyman, Jerzy, D. M. Dabrowska, and T. P. Speed. 1990. “On the Application of Probability Theory to Agricultural Experiments. Essay on Principles. Section 9.” *Statistical Science* 5 (4): 465–72. <https://doi.org/10.1214/ss/1177012031>.
- Vigen, Tyler. 2024. “Tyler Vigen Spurious Correlation #1,172: Butter Consumption Correlates with Economic Output of Washington Metro Area.” TylerVigen.com. January 2024. https://tylervigen.com/spurious/correlation/1172_butter-consumption_correlates-with_economic-output-of-washington-metro-area.